# Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.14)

## Metric

$$ds^2 = -\frac{e^{(-2\,t)}}{z}\mathrm{d}t \otimes \mathrm{d}t + \frac{H\left(z\right)e^{(-2\,t)}}{z}\mathrm{d}t \otimes \mathrm{d}y + 2\,e^{(2\,y)}\mathrm{d}x \otimes \mathrm{d}x + \frac{H\left(z\right)e^{(-2\,t)}}{z}\mathrm{d}y \otimes \mathrm{d}t + \frac{\left(8\,z^2B\left(z\right)^2 - 3\,B\left(z\right)^2H\left(z\right)^2 + 3\right)e^{(-2\,t)}}{3\,zB\left(z\right)^2}\mathrm{d}y \otimes \mathrm{d}y + \frac{e^{(-2\,t)}}{2\,z^2A\left(z\right)B\left(z\right)}\mathrm{d}y \otimes \mathrm{d}z + \frac{e^{(-2\,t)}}{2\,z^2A\left(z\right)B\left(z\right)}\mathrm{d}z \otimes \mathrm{d}y + \frac{e^{(-2\,t)}}{4\,z^3A\left(z\right)^2}\mathrm{d}z \otimes \mathrm{d}z$$

## Coordinates

$(x^\mu) = (t, x, y, z)$

## Metric Functions

$$A = \frac{\sqrt{\frac{1}{2}}\sqrt{4\,z^2 + 4\,z + 3}}{2\,\sqrt{z}}$$

$$B = \frac{\sqrt{3}}{2\,\sqrt{z}}$$

$$H = \frac{2\,A\left(z\right)}{B\left(z\right)} + 1$$

## Null Tetrad

*Covariant components*

$$l = \left(\frac{\sqrt{2}e^{(-t)}}{2\,\sqrt{z}}\right)\,dt + \left(-\frac{\sqrt{2}H\left(z\right)e^{(-t)}}{2\,\sqrt{z}} + \frac{\sqrt{2}e^{(-t)}}{2\,\sqrt{z}B\left(z\right)}\right)\,dy + \left(\frac{\sqrt{2}e^{(-t)}}{4\,z^{\frac{3}{2}}A\left(z\right)}\right)\,dz$$

$$n = \left(\frac{\sqrt{2}e^{(-t)}}{2\sqrt{z}}\right) dt + \left(-\frac{\sqrt{2}H(z)e^{(-t)}}{2\sqrt{z}} - \frac{\sqrt{2}e^{(-t)}}{2\sqrt{z}B(z)}\right) dy + \left(-\frac{\sqrt{2}e^{(-t)}}{4z^{\frac{3}{2}}A(z)}\right) dz$$

$$m = (e^{y})\, dx + \left(2i\sqrt{\frac{1}{3}}\sqrt{z}e^{(-t)}\right) dy$$

$$\overline{m} = (e^{y})\, dx + \left(-2i\sqrt{\frac{1}{3}}\sqrt{z}e^{(-t)}\right) dy$$

*Contravariant components*

$$l = \left(-\frac{1}{2}\sqrt{2}\sqrt{z}e^{t}\right) dt + \left(\sqrt{2}z^{\frac{3}{2}}A(z)e^{t}\right) dz$$

$$n = \left(-\frac{1}{2}\sqrt{2}\sqrt{z}e^{t}\right) dt + \left(-\sqrt{2}z^{\frac{3}{2}}A(z)e^{t}\right) dz$$

$$m = \left(\frac{i\sqrt{3}H(z)e^{t}}{4\sqrt{z}}\right) dt + \left(\frac{1}{2}e^{(-y)}\right) dx + \left(\frac{i\sqrt{3}e^{t}}{4\sqrt{z}}\right) dy + \left(-\frac{i\sqrt{3}\sqrt{z}A(z)e^{t}}{2B(z)}\right) dz$$

$$\overline{m} = \left(-\frac{i\sqrt{3}H(z)e^{t}}{4\sqrt{z}}\right) dt + \left(\frac{1}{2}e^{(-y)}\right) dx + \left(-\frac{i\sqrt{3}e^{t}}{4\sqrt{z}}\right) dy + \left(\frac{i\sqrt{3}\sqrt{z}A(z)e^{t}}{2B(z)}\right) dz$$

## Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\sqrt{z}e^{t}\,\partial_t + \sqrt{2}z^{\frac{3}{2}}A(z)e^{t}\,\partial_z$$

$$\Delta = -\frac{1}{2}\sqrt{2}\sqrt{z}e^{t}\,\partial_t - \sqrt{2}z^{\frac{3}{2}}A(z)e^{t}\,\partial_z$$

$$\delta = \frac{i\sqrt{3}H(z)\,e^{t}}{4\sqrt{z}}\,\partial_t + \frac{1}{2}\,e^{(-y)}\,\partial_x + \frac{i\sqrt{3}e^{t}}{4\sqrt{z}}\,\partial_y - \frac{i\sqrt{3}\sqrt{z}A(z)\,e^{t}}{2\,B(z)}\,\partial_z$$

$$\overline{\delta} = -\frac{i\sqrt{3}H(z)\,e^{t}}{4\sqrt{z}}\,\partial_t + \frac{1}{2}\,e^{(-y)}\,\partial_x - \frac{i\sqrt{3}e^{t}}{4\sqrt{z}}\,\partial_y + \frac{i\sqrt{3}\sqrt{z}A(z)\,e^{t}}{2\,B(z)}\,\partial_z$$

## Spin Coefficients

$$\rho = -\frac{1}{4}\sqrt{2}\sqrt{z}A(z)\,e^{t} - \frac{1}{4}\sqrt{2}\sqrt{z}e^{t}$$

$$\sigma = \frac{1}{4}\sqrt{2}\sqrt{z}A(z)\,e^{t} + \frac{1}{4}\sqrt{2}\sqrt{z}e^{t}$$

$$\kappa = \frac{1}{4}i\sqrt{3}\sqrt{z}A(z)\,e^{t}\frac{\partial}{\partial z}H(z) + \frac{i\sqrt{3}\sqrt{z}A(z)\,e^{t}\frac{\partial}{\partial z}B(z)}{4\,B(z)^{2}}$$

$$\mu = -\frac{1}{4}\sqrt{2}\sqrt{z}A(z)\,e^{t} + \frac{1}{4}\sqrt{2}\sqrt{z}e^{t}$$

$$\lambda = \frac{1}{4}\sqrt{2}\sqrt{z}A(z)\,e^{t} - \frac{1}{4}\sqrt{2}\sqrt{z}e^{t}$$

$$\nu = -\frac{1}{4}i\sqrt{3}\sqrt{z}A(z)\,e^{t}\frac{\partial}{\partial z}H(z) + \frac{i\sqrt{3}\sqrt{z}A(z)\,e^{t}\frac{\partial}{\partial z}B(z)}{4\,B(z)^{2}}$$

$$\alpha = -\frac{1}{8}i\sqrt{3}\sqrt{z}A(z)\,e^{t}\frac{\partial}{\partial z}H(z) + \frac{i\sqrt{3}e^{t}}{8\sqrt{z}}$$

$$\beta = \frac{1}{8}i\sqrt{3}\sqrt{z}A(z)\,e^{t}\frac{\partial}{\partial z}H(z) + \frac{i\sqrt{3}e^{t}}{8\sqrt{z}}$$

$$\tau = \frac{i\sqrt{3}H(z)e^{t}}{4\sqrt{z}} - \frac{i\sqrt{3}\sqrt{z}A(z)e^{t}\frac{\partial}{\partial z}B(z)}{4B(z)^{2}} - \frac{i\sqrt{3}A(z)e^{t}}{4\sqrt{z}B(z)}$$

$$\pi = \frac{i\sqrt{3}H(z)e^{t}}{4\sqrt{z}} - \frac{i\sqrt{3}\sqrt{z}A(z)e^{t}\frac{\partial}{\partial z}B(z)}{4B(z)^{2}} - \frac{i\sqrt{3}A(z)e^{t}}{4\sqrt{z}B(z)}$$

$$\epsilon = -\frac{1}{4}\sqrt{2}\sqrt{z}A(z)e^{t} + \frac{1}{4}\sqrt{2}\sqrt{z}e^{t}$$

$$\gamma = -\frac{1}{4}\sqrt{2}\sqrt{z}A(z)e^{t} - \frac{1}{4}\sqrt{2}\sqrt{z}e^{t}$$

## Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = {} & \frac{3}{8}zA(z)^{2}e^{(2t)}\frac{\partial}{\partial z}H(z)^{2} - \frac{1}{2}z^{2}A(z)e^{(2t)}\frac{\partial}{\partial z}A(z) - \frac{3}{4}zA(z)^{2}e^{(2t)} + \frac{3}{16}A(z)H(z)e^{(2t)}\frac{\partial}{\partial z}H(z) + \frac{3zA(z)e^{(2t)}\frac{\partial}{\partial z}A(z)\frac{\partial}{\partial z}H(z)}{8B(z)} \\ & + \frac{3zA(z)^{2}e^{(2t)}\frac{\partial^{2}}{(\partial z)^{2}}H(z)}{8B(z)} - \frac{1}{2}zA(z)e^{(2t)} - \frac{3}{16}A(z)e^{(2t)}\frac{\partial}{\partial z}H(z) - \frac{3A(z)^{2}e^{(2t)}\frac{\partial}{\partial z}H(z)}{16B(z)} + \frac{1}{4}ze^{(2t)} + \frac{3A(z)H(z)e^{(2t)}\frac{\partial}{\partial z}B(z)}{16B(z)^{2}} \\ & + \frac{3zA(z)e^{(2t)}\frac{\partial}{\partial z}A(z)\frac{\partial}{\partial z}B(z)}{8B(z)^{3}} - \frac{9zA(z)^{2}e^{(2t)}\frac{\partial}{\partial z}B(z)^{2}}{8B(z)^{4}} + \frac{3zA(z)^{2}e^{(2t)}\frac{\partial^{2}}{(\partial z)^{2}}B(z)}{8B(z)^{3}} - \frac{3A(z)^{2}e^{(2t)}\frac{\partial}{\partial z}B(z)}{16B(z)^{3}} - \frac{3A(z)e^{(2t)}\frac{\partial}{\partial z}B(z)}{16B(z)^{2}}\end{aligned}$$

$$\begin{aligned}\Phi_{01} = {} & \frac{1}{8}i\sqrt{3}\sqrt{2}z^{2}A(z)e^{(2t)}\frac{\partial}{\partial z}A(z)\frac{\partial}{\partial z}H(z) + \frac{1}{8}i\sqrt{3}\sqrt{2}z^{2}A(z)^{2}e^{(2t)}\frac{\partial^{2}}{(\partial z)^{2}}H(z) - \frac{1}{16}i\sqrt{3}\sqrt{2}zA(z)^{2}e^{(2t)}\frac{\partial}{\partial z}H(z) + \frac{1}{16}i\sqrt{3}\sqrt{2}zA(z)e^{(2t)}\frac{\partial}{\partial z}H(z) \\ & - \frac{1}{16}i\sqrt{3}\sqrt{2}A(z)H(z)e^{(2t)} - \frac{i\sqrt{3}\sqrt{2}zA(z)e^{(2t)}\frac{\partial}{\partial z}A(z)}{8B(z)} + \frac{1}{16}i\sqrt{3}\sqrt{2}A(z)e^{(2t)} + \frac{i\sqrt{3}\sqrt{2}A(z)^{2}e^{(2t)}}{16B(z)} - \frac{1}{16}i\sqrt{3}\sqrt{2}H(z)e^{(2t)} + \frac{1}{16}i\sqrt{3}\sqrt{2}e^{(2t)} + \frac{i\sqrt{3}\sqrt{2}A(z)e^{(2t)}}{16B(z)}\end{aligned}$$

$$\begin{aligned}\Phi_{02} = {} & -\frac{3}{16}zA(z)^{2}e^{(2t)}\frac{\partial}{\partial z}H(z)^{2} + \frac{1}{2}z^{2}A(z)e^{(2t)}\frac{\partial}{\partial z}A(z) + \frac{1}{4}zA(z)^{2}e^{(2t)} - \frac{1}{4}ze^{(2t)} - \frac{3A(z)H(z)e^{(2t)}\frac{\partial}{\partial z}B(z)}{16B(z)^{2}} \\ & - \frac{3zA(z)e^{(2t)}\frac{\partial}{\partial z}A(z)\frac{\partial}{\partial z}B(z)}{8B(z)^{3}} + \frac{9zA(z)^{2}e^{(2t)}\frac{\partial}{\partial z}B(z)^{2}}{8B(z)^{4}} - \frac{3zA(z)^{2}e^{(2t)}\frac{\partial^{2}}{(\partial z)^{2}}B(z)}{8B(z)^{3}} + \frac{3A(z)e^{(2t)}\frac{\partial}{\partial z}H(z)}{8B(z)} + \frac{3H(z)e^{(2t)}}{16z} \\ & - \frac{3A(z)H(z)e^{(2t)}}{8zB(z)} - \frac{3A(z)e^{(2t)}\frac{\partial}{\partial z}A(z)}{8B(z)^{2}} + \frac{9A(z)^{2}e^{(2t)}\frac{\partial}{\partial z}B(z)}{16B(z)^{3}} - \frac{3A(z)e^{(2t)}\frac{\partial}{\partial z}B(z)}{16B(z)^{2}} + \frac{3A(z)^{2}e^{(2t)}}{8zB(z)^{2}} - \frac{3A(z)e^{(2t)}}{16zB(z)}\end{aligned}$$

$$\Phi_{10} = -\frac{1}{8} i \sqrt{3}\sqrt{2} z^2 A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) \frac{\partial}{\partial z} H(z) - \frac{1}{8} i \sqrt{3}\sqrt{2} z^2 A(z)^2 e^{(2t)} \frac{\partial^2}{(\partial z)^2} H(z) + \frac{1}{16} i \sqrt{3}\sqrt{2} z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} H(z) - \frac{1}{16} i \sqrt{3}\sqrt{2} z A(z) e^{(2t)} \frac{\partial}{\partial z} H(z)$$
$$+ \frac{1}{16} i \sqrt{3}\sqrt{2} A(z) H(z) e^{(2t)} + \frac{i \sqrt{3}\sqrt{2} z A(z) e^{(2t)} \frac{\partial}{\partial z} A(z)}{8 B(z)} - \frac{1}{16} i \sqrt{3}\sqrt{2} A(z) e^{(2t)} - \frac{i \sqrt{3}\sqrt{2} A(z)^2 e^{(2t)}}{16 B(z)} + \frac{1}{16} i \sqrt{3}\sqrt{2} H(z) e^{(2t)} - \frac{1}{16} i \sqrt{3}\sqrt{2} e^{(2t)} - \frac{i \sqrt{3}\sqrt{2} A(z) e^{(2t)}}{16 B(z)}$$

$$\Phi_{11} = \frac{3}{32} z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} H(z)^2 - \frac{1}{2} z^2 A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) + \frac{3 H(z)^2 e^{(2t)}}{32 z} - \frac{3 A(z) H(z) e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^2}$$
$$- \frac{3 H(z) e^{(2t)}}{32 z} - \frac{3 A(z) H(z) e^{(2t)}}{16 z B(z)} + \frac{3 A(z)^2 e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^3} - \frac{3 e^{(2t)}}{32 z} + \frac{3 A(z)^2 e^{(2t)}}{32 z B(z)^2} - \frac{3 A(z) e^{(2t)}}{32 z B(z)}$$

$$\Phi_{12} = \frac{1}{8} i \sqrt{3}\sqrt{2} z^2 A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) \frac{\partial}{\partial z} H(z) + \frac{1}{8} i \sqrt{3}\sqrt{2} z^2 A(z)^2 e^{(2t)} \frac{\partial^2}{(\partial z)^2} H(z) - \frac{1}{16} i \sqrt{3}\sqrt{2} z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} H(z) - \frac{1}{16} i \sqrt{3}\sqrt{2} z A(z) e^{(2t)} \frac{\partial}{\partial z} H(z)$$
$$+ \frac{1}{16} i \sqrt{3}\sqrt{2} A(z) H(z) e^{(2t)} + \frac{i \sqrt{3}\sqrt{2} z A(z) e^{(2t)} \frac{\partial}{\partial z} A(z)}{8 B(z)} - \frac{1}{16} i \sqrt{3}\sqrt{2} A(z) e^{(2t)} - \frac{i \sqrt{3}\sqrt{2} A(z)^2 e^{(2t)}}{16 B(z)} - \frac{1}{16} i \sqrt{3}\sqrt{2} H(z) e^{(2t)} + \frac{1}{16} i \sqrt{3}\sqrt{2} e^{(2t)} + \frac{i \sqrt{3}\sqrt{2} A(z) e^{(2t)}}{16 B(z)}$$

$$\Phi_{20} = -\frac{3}{16} z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} H(z)^2 + \frac{1}{2} z^2 A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) + \frac{1}{4} z A(z)^2 e^{(2t)} - \frac{1}{4} z e^{(2t)} - \frac{3 A(z) H(z) e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^2}$$
$$- \frac{3 z A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) \frac{\partial}{\partial z} B(z)}{8 B(z)^3} + \frac{9 z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} B(z)^2}{8 B(z)^4} - \frac{3 z A(z)^2 e^{(2t)} \frac{\partial^2}{(\partial z)^2} B(z)}{8 B(z)^3} + \frac{3 A(z) e^{(2t)} \frac{\partial}{\partial z} H(z)}{8 B(z)} + \frac{3 H(z) e^{(2t)}}{16 z}$$
$$- \frac{3 A(z) H(z) e^{(2t)}}{8 z B(z)} - \frac{3 A(z) e^{(2t)} \frac{\partial}{\partial z} A(z)}{8 B(z)^2} + \frac{9 A(z)^2 e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^3} - \frac{3 A(z) e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^2} + \frac{3 A(z)^2 e^{(2t)}}{8 z B(z)^2} - \frac{3 A(z) e^{(2t)}}{16 z B(z)}$$

$$\Phi_{21} = -\frac{1}{8} i \sqrt{3}\sqrt{2} z^2 A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) \frac{\partial}{\partial z} H(z) - \frac{1}{8} i \sqrt{3}\sqrt{2} z^2 A(z)^2 e^{(2t)} \frac{\partial^2}{(\partial z)^2} H(z) + \frac{1}{16} i \sqrt{3}\sqrt{2} z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} H(z) + \frac{1}{16} i \sqrt{3}\sqrt{2} z A(z) e^{(2t)} \frac{\partial}{\partial z} H(z)$$
$$- \frac{1}{16} i \sqrt{3}\sqrt{2} A(z) H(z) e^{(2t)} - \frac{i \sqrt{3}\sqrt{2} z A(z) e^{(2t)} \frac{\partial}{\partial z} A(z)}{8 B(z)} + \frac{1}{16} i \sqrt{3}\sqrt{2} A(z) e^{(2t)} + \frac{i \sqrt{3}\sqrt{2} A(z)^2 e^{(2t)}}{16 B(z)} + \frac{1}{16} i \sqrt{3}\sqrt{2} H(z) e^{(2t)} - \frac{1}{16} i \sqrt{3}\sqrt{2} e^{(2t)} - \frac{i \sqrt{3}\sqrt{2} A(z) e^{(2t)}}{16 B(z)}$$

$$\Phi_{22} = \frac{3}{8} z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} H(z)^2 - \frac{1}{2} z^2 A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) - \frac{3}{4} z A(z)^2 e^{(2t)} - \frac{3}{16} A(z) H(z) e^{(2t)} \frac{\partial}{\partial z} H(z) - \frac{3 z A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) \frac{\partial}{\partial z} H(z)}{8 B(z)}$$
$$- \frac{3 z A(z)^2 e^{(2t)} \frac{\partial^2}{(\partial z)^2} H(z)}{8 B(z)} + \frac{1}{2} z A(z) e^{(2t)} + \frac{3}{16} A(z) e^{(2t)} \frac{\partial}{\partial z} H(z) + \frac{3 A(z)^2 e^{(2t)} \frac{\partial}{\partial z} H(z)}{16 B(z)} + \frac{1}{4} z e^{(2t)} + \frac{3 A(z) H(z) e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^2}$$
$$+ \frac{3 z A(z) e^{(2t)} \frac{\partial}{\partial z} A(z) \frac{\partial}{\partial z} B(z)}{8 B(z)^3} - \frac{9 z A(z)^2 e^{(2t)} \frac{\partial}{\partial z} B(z)^2}{8 B(z)^4} + \frac{3 z A(z)^2 e^{(2t)} \frac{\partial^2}{(\partial z)^2} B(z)}{8 B(z)^3} - \frac{3 A(z)^2 e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^3} - \frac{3 A(z) e^{(2t)} \frac{\partial}{\partial z} B(z)}{16 B(z)^2}$$

$$\Lambda = \frac{1}{32}\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)^{2} - \frac{1}{12}\, zA\left(z\right)^{2} e^{\left(2\,t\right)} + \frac{1}{12}\, ze^{\left(2\,t\right)} - \frac{H\left(z\right)^{2} e^{\left(2\,t\right)}}{32\, z} + \frac{A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{8\, B\left(z\right)^{2}} + \frac{zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} B\left(z\right)}{8\, B\left(z\right)^{3}}$$
$$- \frac{3\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)^{2}}{8\, B\left(z\right)^{4}} + \frac{zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial^{2}}{\left(\partial z\right)^{2}} B\left(z\right)}{8\, B\left(z\right)^{3}} - \frac{A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)}{8\, B\left(z\right)} + \frac{H\left(z\right) e^{\left(2\,t\right)}}{32\, z} + \frac{3\, A\left(z\right) H\left(z\right) e^{\left(2\,t\right)}}{16\, zB\left(z\right)}$$
$$+ \frac{A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right)}{8\, B\left(z\right)^{2}} - \frac{A\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{4\, B\left(z\right)^{3}} - \frac{A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^{2}} - \frac{e^{\left(2\,t\right)}}{32\, z} - \frac{5\, A\left(z\right)^{2} e^{\left(2\,t\right)}}{32\, zB\left(z\right)^{2}} - \frac{3\, A\left(z\right) e^{\left(2\,t\right)}}{32\, zB\left(z\right)}$$

## Weyl Tensor Components

$$\Psi_{0} = -\frac{3}{8}\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)^{2} + \frac{1}{2}\, z^{2} A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) + \frac{3}{4}\, zA\left(z\right)^{2} e^{\left(2\,t\right)} - \frac{3}{16}\, A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right) - \frac{3\, zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} H\left(z\right)}{8\, B\left(z\right)}$$
$$- \frac{3\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial^{2}}{\left(\partial z\right)^{2}} H\left(z\right)}{8\, B\left(z\right)} + \frac{1}{2}\, zA\left(z\right) e^{\left(2\,t\right)} - \frac{3}{16}\, A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right) + \frac{3\, A\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)}{16\, B\left(z\right)} - \frac{1}{4}\, ze^{\left(2\,t\right)} - \frac{3\, A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^{2}}$$
$$- \frac{3\, zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} B\left(z\right)}{8\, B\left(z\right)^{3}} + \frac{9\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)^{2}}{8\, B\left(z\right)^{4}} - \frac{3\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial^{2}}{\left(\partial z\right)^{2}} B\left(z\right)}{8\, B\left(z\right)^{3}} + \frac{3\, A\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^{3}} - \frac{3\, A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^{2}}$$

$$\Psi_{1} = \frac{1}{8} i\, \sqrt{3}\sqrt{2} z^{2} A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} H\left(z\right) + \frac{1}{8} i\, \sqrt{3}\sqrt{2} z^{2} A\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial^{2}}{\left(\partial z\right)^{2}} H\left(z\right) - \frac{1}{16} i\, \sqrt{3}\sqrt{2} zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right) + \frac{1}{16} i\, \sqrt{3}\sqrt{2} zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)$$
$$- \frac{1}{16} i\, \sqrt{3}\sqrt{2} A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} - \frac{i\, \sqrt{3}\sqrt{2} zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right)}{8\, B\left(z\right)} - \frac{1}{16} i\, \sqrt{3}\sqrt{2} A\left(z\right) e^{\left(2\,t\right)} + \frac{i\, \sqrt{3}\sqrt{2} A\left(z\right)^{2} e^{\left(2\,t\right)}}{16\, B\left(z\right)} - \frac{1}{16} i\, \sqrt{3}\sqrt{2} H\left(z\right) e^{\left(2\,t\right)} - \frac{1}{16} i\, \sqrt{3}\sqrt{2} e^{\left(2\,t\right)} + \frac{i\, \sqrt{3}\sqrt{2} A\left(z\right) e^{\left(2\,t\right)}}{16\, B\left(z\right)}$$

$$\Psi_{2} = \frac{1}{8}\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)^{2} - \frac{1}{2}\, z^{2} A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) - \frac{1}{12}\, zA\left(z\right)^{2} e^{\left(2\,t\right)} + \frac{1}{12}\, ze^{\left(2\,t\right)} + \frac{H\left(z\right)^{2} e^{\left(2\,t\right)}}{16\, z} - \frac{A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^{2}} + \frac{zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} B\left(z\right)}{8\, B\left(z\right)^{3}}$$
$$- \frac{3\, zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)^{2}}{8\, B\left(z\right)^{4}} + \frac{zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial^{2}}{\left(\partial z\right)^{2}} B\left(z\right)}{8\, B\left(z\right)^{3}} - \frac{A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)}{8\, B\left(z\right)} + \frac{H\left(z\right) e^{\left(2\,t\right)}}{8\, z} + \frac{A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right)}{8\, B\left(z\right)^{2}} - \frac{A\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^{3}} - \frac{A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^{2}} + \frac{e^{\left(2\,t\right)}}{16\, z} - \frac{A\left(z\right)^{2} e^{\left(2\,t\right)}}{16\, zB\left(z\right)^{2}}$$

$$\Psi_{3} = -\frac{1}{8} i\, \sqrt{3}\sqrt{2} z^{2} A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} H\left(z\right) - \frac{1}{8} i\, \sqrt{3}\sqrt{2} z^{2} A\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial^{2}}{\left(\partial z\right)^{2}} H\left(z\right) + \frac{1}{16} i\, \sqrt{3}\sqrt{2} zA\left(z\right)^{2} e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right) + \frac{1}{16} i\, \sqrt{3}\sqrt{2} zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)$$
$$- \frac{1}{16} i\, \sqrt{3}\sqrt{2} A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} - \frac{i\, \sqrt{3}\sqrt{2} zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right)}{8\, B\left(z\right)} - \frac{1}{16} i\, \sqrt{3}\sqrt{2} A\left(z\right) e^{\left(2\,t\right)} + \frac{i\, \sqrt{3}\sqrt{2} A\left(z\right)^{2} e^{\left(2\,t\right)}}{16\, B\left(z\right)} + \frac{1}{16} i\, \sqrt{3}\sqrt{2} H\left(z\right) e^{\left(2\,t\right)} + \frac{1}{16} i\, \sqrt{3}\sqrt{2} e^{\left(2\,t\right)} - \frac{i\, \sqrt{3}\sqrt{2} A\left(z\right) e^{\left(2\,t\right)}}{16\, B\left(z\right)}$$

$$\Psi_4 = -\frac{3}{8}\, zA\left(z\right)^2 e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)^2 + \frac{1}{2}\, z^2 A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) + \frac{3}{4}\, zA\left(z\right)^2 e^{\left(2\,t\right)} + \frac{3}{16}\, A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right) + \frac{3\, zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} H\left(z\right)}{8\, B\left(z\right)}$$
$$+ \frac{3\, zA\left(z\right)^2 e^{\left(2\,t\right)} \frac{\partial^2}{\left(\partial z\right)^2} H\left(z\right)}{8\, B\left(z\right)} - \frac{1}{2}\, zA\left(z\right) e^{\left(2\,t\right)} + \frac{3}{16}\, A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right) - \frac{3\, A\left(z\right)^2 e^{\left(2\,t\right)} \frac{\partial}{\partial z} H\left(z\right)}{16\, B\left(z\right)} - \frac{1}{4}\, ze^{\left(2\,t\right)} - \frac{3\, A\left(z\right) H\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^2}$$
$$- \frac{3\, zA\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} A\left(z\right) \frac{\partial}{\partial z} B\left(z\right)}{8\, B\left(z\right)^3} + \frac{9\, zA\left(z\right)^2 e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)^2}{8\, B\left(z\right)^4} - \frac{3\, zA\left(z\right)^2 e^{\left(2\,t\right)} \frac{\partial^2}{\left(\partial z\right)^2} B\left(z\right)}{8\, B\left(z\right)^3} + \frac{3\, A\left(z\right)^2 e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^3} - \frac{3\, A\left(z\right) e^{\left(2\,t\right)} \frac{\partial}{\partial z} B\left(z\right)}{16\, B\left(z\right)^2}$$

## Newman-Penrose Equations

*Equations are vanished.*

## Bianchi Identities

*Equations are vanished.*

## Petrov Type

Type "$I$"